AP CALCULUS AB

Unit 1 • Lesson 1.1 notes

Lesson 1.1 • Introducing Calculus: Can Change Occur at an Instant?

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Explain why an instantaneous rate of change cannot be computed with a single average rate of change.
- Estimate an instantaneous rate of change from average rates of change over shrinking intervals.

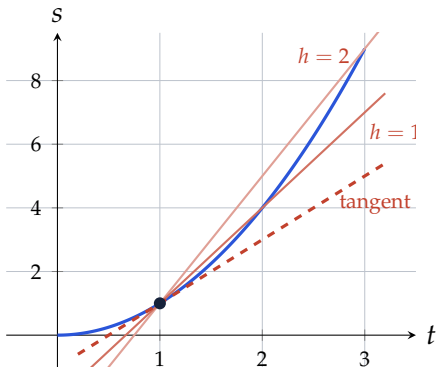
Vocabulary. average rate of change, instantaneous rate of change, secant line, tangent line, limit.

1. The problem of the instant

A speedometer shows a speed *at an instant*, yet speed is distance divided by time, and in a single instant no time passes and no distance is covered: $\frac{0}{0}$. Calculus resolves this by looking at what average rates of change *approach* as the time interval shrinks.

2. From secant lines to the tangent line

The average rate of change of f on $[a, a+h]$ is $\dfrac{f(a+h)-f(a)}{h}$, the slope of a secant line. As $h \to 0$ the secant lines approach the **tangent line** at $x = a$, and their slopes approach the instantaneous rate of change.



Secant lines of $s(t) = t^2$ from $t = 1$ with slopes 4 and 3 approach the tangent line of slope 2.

Example 1 • Algebra

For $s(t) = t^2$, find the average velocity on $[1, 1+h]$ and the velocity at $t = 1$.

Solution. $\dfrac{(1+h)^2 - 1}{h} = \dfrac{2h + h^2}{h} = 2 + h$ for $h \neq 0$. As $h \to 0$ this approaches 2: the velocity at $t = 1$ is 2.

Example 2 • Table

h	0.1	0.01	−0.01	−0.1
$\frac{f(2+h)-f(2)}{h}$	12.61	12.0601	11.9401	11.41

Solution. The values approach 12 from both sides, so the instantaneous rate of change of f at $x = 2$ is about 12 (here $f(x) = x^3$).

Common mistakes

- Substituting $h = 0$ into $\frac{f(a+h)-f(a)}{h}$ before simplifying, which gives $\frac{0}{0}$.
- Using only positive h: check that values from both sides agree.

Quick check

1. Simplify $\dfrac{(3+h)^2 - 9}{h}$ and find what it approaches as $h \to 0$.
2. What is the instantaneous rate of change of $f(x) = 7 - 2x$ at any x?

Answers. 1. $6 + h \to 6$. 2. -2, the slope.